AI Smart Farming Assistant

Final Year Project Proposal Summary — The Islamia University of Bahawalpur

Project Overview

AI Smart Farming Assistant is a web-based application that helps farmers improve crop health and productivity using Artificial Intelligence. It gives farmers quick and simple advice so they can make better farming decisions.

Problem

Farmers often find it hard to identify crop diseases, manage irrigation, and choose suitable fertilizers. Traditional methods take time and cost money. This project offers a smart, easy, and low-cost solution.

Main Objectives

- **Disease detection:** identify possible crop diseases from leaf images.
- **Treatment advice:** suggest prevention and treatment methods.
- **Water estimate:** estimate irrigation needs using weather and soil information.
- **Cost calculation:** calculate fertilizer costs and total farming expenses.
- **Two languages:** support both Urdu and English.

Technology Stack

- **Language:** Python
- **Framework:** Flask
- **Database:** SQLite or MySQL
- **Tools:** VS Code and a web browser
- **Architecture:** Three-tier web architecture (presentation, logic, and data layers)

Expected Outcome

A simple, affordable web application that helps even small-scale farmers use modern technology to grow healthier crops and increase their income.

Project Details

- **Student:** Khan Mohammad (F23BINFT1E02143)
- **Program:** BSIT
- **Supervisor:** Dost Muhammad Khan